

Project Design Phase-II

Data Flow Diagram & User Stories

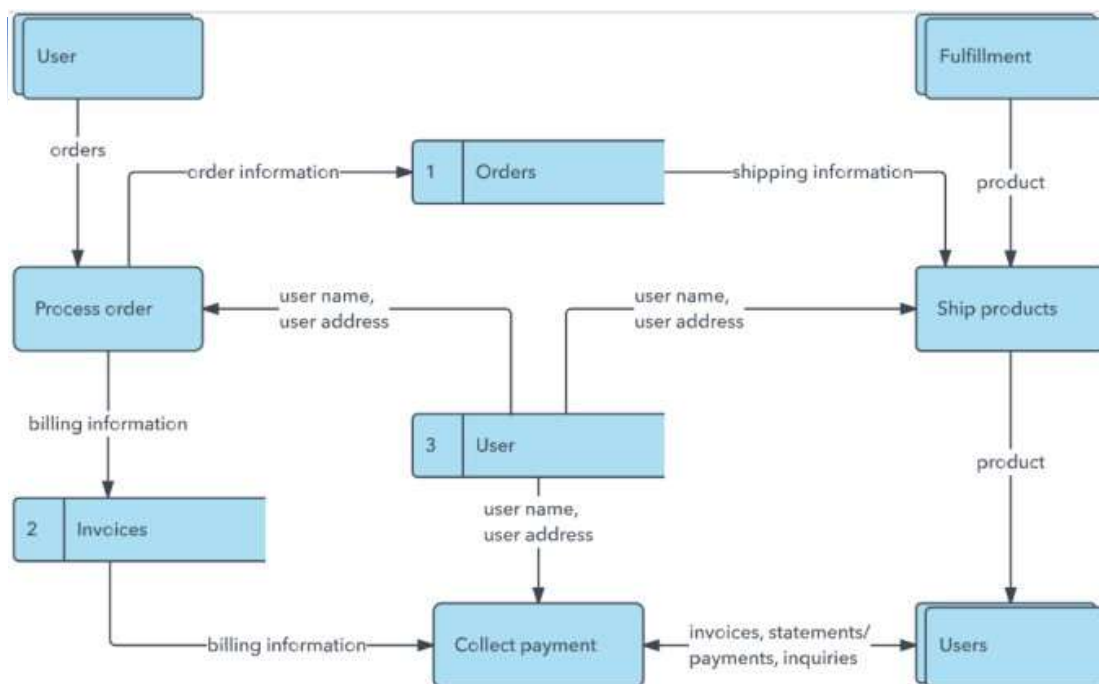
Date	23 September 2026
Team ID	SWTID-2026-9721
Project Name	AI WEATHERWISE API
Maximum Marks	4 Marks

Data Flow Diagrams:

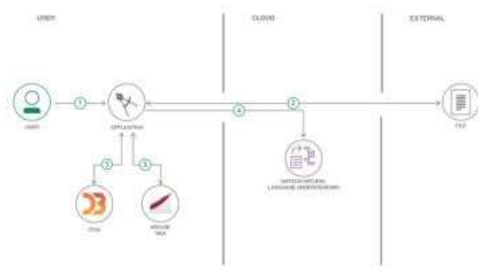
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: DFD Level 0 (Industry Standard)

Example: [Simplified](#)



Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
Customer	Registration	USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
Customer	Registration	USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer	Registration	USN-4	As a user, I can register for the application through Gmail	I can register successfully using a valid mobile number.	Medium	Sprint-1
Customer	Login	USN-5	As a user, I can log into the application by entering email & password	I can successfully log in with valid credentials.	High	Sprint-1
Customer	Dashboard	USN-6	As a user, I can access the Weather Wise dashboard after logging in.	I can access the dashboard successfully after login.	High	Sprint-1
Customer (Web user)	Weather Information	USN-7	As a user, I can view the current weather information for a selected location.	I can view the current weather details for the selected location.	High	Sprint-2
Customer Care Executive	Weather Forecast	USN-8	As a user, I can view the weather forecast for upcoming days.	I can view the current weather forecast for the selected location.	High	Sprint-2
Administrator	Weather Recommendation	USN-9	As a user, I can receive recommendations based on the current weather conditions.	I can view appropriate recommendations based on the weather data.	Medium	Sprint-2
Customer	Logout	USN-10	As a user, I can securely log out from the Weather Wise application.	I am successfully logged out and redirected to the login page.	Medium	Sprint-3

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIN
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	User Login	Login through Email & Password Login through Gmail
FR-4	Dashboard	View User Dashboard Login through LinkedIn View User Account Information

Access Dashboard after Login

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should be simple, user-friendly and easy to navigate.
NFR-2	Security	User credentials and personal information should be securely protected.

NFR-3	Reliability	The application should perform consistently without frequent failures.
NFR-4	Performance	The application should respond quickly to user requests.
NFR-5	Availability	The application should be available to users whenever required.
NFR-6	Scalability	The application should support an increasing number of users without major performance issues.

Project Design Phase-II

Technology Stack (Architecture & Stack)

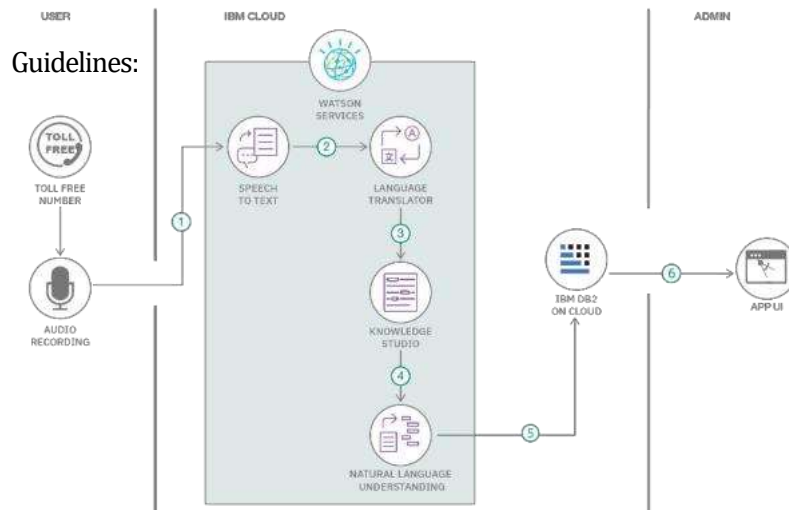
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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Include all
the processes
(As an
application
logic /
Technology
Block)

Provide
infrastructural
demarcation
(Local /
Cloud)

Indicate
external
interfaces (third
party API's etc.)

Indicate
Data Storage
components /
services

Indicate
interface to
machine learning
models (if
applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>